



Chronic Kidney Disease

(Chronic Renal Failure; CKD)

By [Anna Malkina](#), MD, University of California, San Francisco

Reviewed By [Navin Jaipaul](#), MD, MHS, Loma Linda University School of Medicine

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Chronic kidney disease (CKD) is long-standing, progressive deterioration of renal function. Symptoms develop slowly and in advanced stages include anorexia, nausea, vomiting, stomatitis, dysgeusia, nocturia, lassitude, fatigue, pruritus, decreased mental acuity, muscle twitches and cramps, water retention, undernutrition, peripheral neuropathies, and seizures. Diagnosis is based on laboratory testing of renal function, sometimes followed by renal biopsy. Treatment is primarily directed at the underlying condition but includes fluid and electrolyte management, blood pressure control, treatment of anemia, various types of dialysis, and kidney transplantation.

Prevalence of CKD (defined as estimated glomerular filtration rate [eGFR] <60 mL/min/1.73m² or urinary albumin to creatinine ratio [ACR] ≥ 30 mg/g) among adults in the United States from 2017 through March 2020 was estimated to be 14.0% (1).

General reference

1. [United States Renal Data System \(USRDS\)](#): CKD in the general population. 2022 Annual Data Update. Accessed January 28, 2025.

Etiology of Chronic Kidney Disease

Chronic kidney disease may result from any cause of renal dysfunction of sufficient magnitude (see table [Major Causes of Chronic Kidney Disease](#)).

The **most common causes** in the United States in order of prevalence are

- [Diabetic nephropathy](#)
- [Hypertensive nephrosclerosis](#)
- Various primary and secondary [glomerulopathies](#)

[Metabolic syndrome](#), in which hypertension and type 2 diabetes are present, is a large and growing cause of renal damage.

TABLE

Major Causes of Chronic Kidney Disease

Cause	Examples
Chronic tubulointerstitial nephropathies	See table Causes of Chronic Tubulointerstitial Nephritis
Glomerulopathies (primary)	Focal segmental glomerulosclerosis Idiopathic crescentic glomerulonephritis IgA nephropathy Membranoproliferative glomerulonephritis Membranous nephropathy
Glomerulopathies associated with systemic disease	Amyloidosis Diabetes mellitus Antiglomerular basement membrane (GBM) antibody disease (also known as Goodpasture syndrome) Granulomatosis with polyangiitis (formerly known as Wegener's granulomatosis) Hemolytic-uremic syndrome Mixed cryoglobulinemia Postinfectious glomerulonephritis Systemic lupus erythematosus
Hereditary nephropathies	Autosomal dominant interstitial kidney disease (medullary cystic kidney disease) Hereditary nephritis (Alport syndrome) Nail-patella syndrome Polycystic kidney disease
Hypertension	Hypertensive nephrosclerosis

Cause	Examples
<u>Obstructive uropathy</u>	<u>Benign prostatic hyperplasia</u>
	<u>Posterior urethral valves</u>
	Retroperitoneal fibrosis
	Ureteral obstruction (congenital, calculi, cancer)
	<u>Vesicoureteral reflux</u>
Renal macrovascular disease (vasculopathy of renal arteries and veins)	<u>Renal artery stenosis</u> caused by atherosclerosis or <u>fibromuscular dysplasia</u>

Pathophysiology of Chronic Kidney Disease

Chronic kidney disease (CKD) is initially described as diminished renal reserve or renal insufficiency, which may progress to renal failure (end-stage kidney disease). Initially, as renal tissue loses function, there are few noticeable abnormalities because the remaining tissue increases its performance (renal functional adaptation).

Decreased renal function interferes with the kidneys' ability to maintain fluid and electrolyte homeostasis. The ability to concentrate urine declines early and is followed by decreases in ability to excrete excess phosphate, acid, and potassium. When renal failure is advanced (glomerular filtration rate [GFR] ≤ 15 mL/min/1.73 m²), the ability to effectively dilute or concentrate urine is lost; thus, urine osmolality is usually fixed at approximately 300 to 320 mOsm/kg, close to that of plasma (275 to 295 mOsm/kg), and urinary volume does not respond readily to variations in water intake.

Creatinine and urea

Plasma concentrations of creatinine and urea (which are highly dependent on glomerular filtration) begin a hyperbolic rise as GFR diminishes. These changes are minimal early on. When the GFR falls below 15 mL/min/1.73 m² (normal > 90 mL/min/1.73 m²), creatinine and urea levels are high and are usually associated with systemic manifestations (uremia). Urea and creatinine are not major contributors to the uremic symptoms; they are markers for many other substances (some not yet well defined) that cause the symptoms.

Sodium and water

Despite a diminishing GFR, sodium and water balance is well-maintained by increased fractional excretion of sodium in urine and a normal response to thirst. Thus, the plasma sodium concentration is typically normal, and hypervolemia is infrequent unless dietary intake of sodium or water is very

restricted or excessive. [Heart failure](#) can occur due to sodium and water overload, particularly in patients with decreased cardiac reserve.

Potassium

For substances whose secretion is controlled mainly through distal nephron secretion (eg, potassium), renal adaptation usually maintains plasma levels at normal until renal failure is advanced or dietary potassium intake is excessive. [Potassium-sparing diuretics](#), [angiotensin-converting enzyme inhibitors](#), [beta-blockers](#), [nonsteroidal anti-inflammatory drugs](#) (NSAIDs), cyclosporine, tacrolimus, trimethoprim/sulfamethoxazole, pentamidine, or [angiotensin II receptor blockers](#) may raise plasma potassium levels in patients with less advanced renal failure.

Calcium and phosphate

Abnormalities of calcium, phosphate, parathyroid hormone (PTH), and [vitamin D metabolism](#) can occur, as can renal osteodystrophy. Decreased renal production of calcitriol (1,25(OH)₂D, the active vitamin D hormone) contributes to [hypocalcemia](#). Decreased renal excretion of phosphate results in [hyperphosphatemia](#). Secondary hyperparathyroidism is common and can develop in renal failure before abnormalities in calcium or phosphate concentrations occur. For this reason, monitoring PTH in patients with moderate CKD, even before hyperphosphatemia occurs, has been recommended.

Renal osteodystrophy (abnormal bone mineralization resulting from hyperparathyroidism, calcitriol deficiency, elevated serum phosphate, or low or normal serum calcium) usually takes the form of increased bone turnover due to hyperparathyroid bone disease (osteitis fibrosa) but can also involve decreased bone turnover due to adynamic bone disease (due to excessive parathyroid suppression) or osteomalacia. Calcitriol deficiency may cause osteopenia or osteomalacia.

pH and bicarbonate

Moderate [metabolic acidosis](#) (plasma bicarbonate content 15 to 20 mmol/L) is characteristic. Acidosis causes muscle wasting due to protein catabolism, bone loss due to bone buffering of acid, and accelerated progression of kidney disease.

Anemia

Anemia is characteristic of moderate to advanced CKD (≥ stage 3). The anemia of CKD is normochromic-normocytic, with a hematocrit of 20 to 30% (35 to 40% in patients with [polycystic kidney disease](#)). It is usually caused by deficient erythropoietin production due to a reduction of functional renal mass (see page [Anemias Caused by Deficient Erythropoiesis](#)). Other causes include [deficiencies of iron](#), [folate](#), and [vitamin B12](#).

Symptoms and Signs of Chronic Kidney Disease

Patients with mildly diminished renal reserve are asymptomatic. Even patients with mild to moderate renal insufficiency may have no symptoms despite elevated blood urea nitrogen (BUN) and creatinine.

Nocturia is often noted, principally due to failure to concentrate the urine. Lassitude, fatigue, anorexia, and decreased mental acuity often are the earliest manifestations of uremia.

With more severe renal disease (eg, estimated glomerular filtration rate [eGFR] < 15 mL/min/1.73 m²), neuromuscular symptoms may be present, including coarse muscular twitches, [peripheral sensory and motor neuropathies](#), muscle cramps, hyperreflexia, [restless legs syndrome](#), and seizures (usually the result of hypertensive or metabolic encephalopathy).

Anorexia, nausea, vomiting, weight loss, stomatitis, and an unpleasant taste in the mouth are almost uniformly present. The skin may be yellow-brown and/or dry. Occasionally, urea from sweat crystallizes on the skin (uremic frost). Pruritus may be especially uncomfortable. Undernutrition leading to generalized tissue wasting is a prominent feature of chronic uremia.

Uremic Frost

IMAGE



This photo shows scattered deposits of white, friable, crystalline material (arrow) with a frosted appearance on the patient's eyebrows, suggesting uremic frost.

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In advanced CKD, [pericarditis](#) and gastrointestinal ulceration and bleeding may occur. Hypertension is present in > 80% of patients with advanced CKD and is usually related to hypervolemia. Heart failure caused by [hypertension](#) or [coronary artery disease](#) and renal retention of sodium and water may lead to dependent edema and/or dyspnea.

Diagnosis of Chronic Kidney Disease

- Estimated glomerular filtration rate (eGFR) using creatinine, or if cystatin C is available then eGFR calculation based on combination of creatinine and cystatin C (1)

- Electrolytes, blood urea nitrogen (BUN), creatinine, phosphate, calcium, complete blood count (CBC)
- Urinalysis (including urinary sediment examination)
- Quantitative urine protein (24-hour urine protein collection or spot urine protein to creatinine ratio)
- Ultrasound
- Sometimes renal biopsy

Chronic kidney disease (CKD) is usually first suspected when serum creatinine rises. The initial step is to determine whether the renal failure is acute, chronic, or acute superimposed on chronic (ie, an acute disease that further compromises renal function in a patient with CKD—see table [Distinguishing Acute Kidney Injury From Chronic Kidney Disease](#)). The cause of renal failure is also determined. Sometimes determining the duration of renal failure helps determine the cause; sometimes it is easier to determine the cause than the duration, and determining the cause helps determine the duration.

TABLE

Distinguishing Acute Kidney Injury From Chronic Kidney Disease

Finding	Comment
Decreased kidney function (eGFR < 60 mL/min/1.73 m ²) for ≥ 3 months	Most reliable evidence of CKD
Renal sonogram showing small kidneys	Usually CKD
Renal sonogram showing normal or enlarged kidneys	May be AKI or some forms of CKD (diabetic nephropathy , acute hypertensive nephrosclerosis , polycystic kidney disease , myeloma , rapidly progressive glomerulonephritis, infiltrative diseases [eg, lymphoma , leukemia , amyloidosis], obstruction)
Oliguria, daily increases in serum creatinine and BUN	Probably AKI or AKI superimposed on CKD
No anemia	Probably AKI or CKD due to polycystic kidney disease
Severe anemia, hyperphosphatemia , and hypocalcemia	Possibly CKD but may be AKI
Subperiosteal erosions on radiography	Probably CKD
Chronic symptoms or signs (eg, fatigue, nausea, pruritus, nocturia, hypertension)	Usually CKD

AKI = acute kidney injury; CKD = chronic kidney disease; BUN = blood urea nitrogen; eGFR = estimated glomerular filtration rate

The etiology of acute kidney injury (AKI) and CKD is determined based on clinical context, personal and family history, social and environmental factors, medications (including over-the-counter) and supplements, physical examination, and possibly genetic testing (1). Sometimes specific serologic tests are needed to determine the cause. In AKI, serum creatinine rises rapidly over days to few months. Urinalysis findings depend on the nature of the underlying disorder, but broad (> 3 white blood cell diameters wide) or especially waxy (highly refractile) casts often are prominent in advanced renal failure of any cause.

An ultrasound examination of the kidneys is usually helpful in evaluating for [obstructive uropathy](#) and in distinguishing acute kidney injury from CKD based on kidney size. Except in certain conditions (see table [Distinguishing Acute Kidney Injury From Chronic Kidney Disease](#)), patients with CKD have small, shrunken kidneys (usually < 10 cm in length) with thinned, hyperechoic cortex. Obtaining a precise diagnosis becomes increasingly difficult as renal function reaches values close to those of end-stage kidney disease. The definitive diagnostic tool is [renal biopsy](#), but it is not recommended when ultrasound indicates small, fibrotic kidneys; high procedural risk outweighs low diagnostic yield.

Stages of chronic kidney disease

Staging CKD is a way of quantifying its severity. CKD has been classified into 5 stages.

- Stage 1: Normal GFR (≥ 90 mL/min/1.73 m²) plus either persistent albuminuria or known structural or hereditary renal disease
- Stage 2: GFR 60 to 89 mL/min/1.73 m²
- Stage 3a: 45 to 59 mL/min/1.73 m²
- Stage 3b: 30 to 44 mL/min/1.73 m²
- Stage 4: GFR 15 to 29 mL/min/1.73 m²
- Stage 5: GFR < 15 mL/min/1.73 m²

GFR (in mL/min/1.73 m²) in CKD can be estimated by the Chronic Kidney Disease Epidemiology Collaboration ([CKD-EPI 2021](#)) [creatinine equation](#):

$$eGFR = 142 \times \min\left(\frac{S_{Cr}}{k}, 1\right)^{\alpha} \times \max\left(\frac{S_{Cr}}{k}, 1\right)^{-1.200} \times 0.9938^{age} \times 1.012[if\ female]$$

where

- Scr = standardized serum creatinine in mg/dL
- κ = 0.7 (females) or 0.9 (males)
- α = -0.241 (female) or -0.302 (male)

- $\min(\text{Scr}/\kappa, 1) = \text{minimum of Scr}/\kappa \text{ or } 1.0$
- $\max(\text{Scr}/\kappa, 1) = \text{maximum of Scr}/\kappa \text{ or } 1.0$
- Age (years)

In contrast to prior versions, the most recent equation does not adjust for race, so as to reduce racial inequities in CKD diagnosis and thus treatment. Alternatively, GFR can be estimated using timed (most commonly 24-hour) urine creatinine clearance that includes measured serum and urine creatinine; this equation tends to overestimate GFR by 10 to 20%. It is used when serum creatinine assessment may not be as accurate (eg, in patients who are sedentary, very obese, or very thin). Serum cystatin C is an alternative endogenous GFR marker used as a confirmatory test in people with nonrenal factors affecting serum creatinine level (eg, extremely high or low muscle mass, exogenous creatine intake, amputations or neuromuscular diseases, and high protein or exclusively plant-based diets). GFR is calculated using [CKD-EPI cystatin C equation](#).

The [CKD-EPI 2021 formula](#) is more accurate than the Modification of Diet in Renal Disease (MDRD) and Cockcroft-Gault formulas, particularly for patients with a GFR near normal values. The CKD-EPI equation yields fewer falsely positive results indicating chronic kidney disease and predicts outcome better than the other formulas.

CLINICAL CALCULATORS

[Creatinine Clearance \(measured\)](#)



Diagnosis reference

1. [Kidney Disease: Improving Global Outcomes \(KDIGO\) CKD Work Group](#). KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. *Kidney Int*. 2024;105(4S):S117-S314. doi:10.1016/j.kint.2023.10.018

Treatment of Chronic Kidney Disease

- Control of underlying disorders
- Possible restriction of dietary protein, phosphate, and potassium
- Vitamin D supplements
- Treatment of anemia
- Treatment of contributing comorbidities (eg, heart failure, diabetes mellitus, nephrolithiasis, prostatic hypertrophy)
- Doses of all medications adjusted as needed
- Treatment of metabolic acidosis to a serum bicarbonate goal of >18 mmol/L (>18 mEq/L)
- Dialysis for severely decreased glomerular filtration rate (GFR) if symptoms and signs not adequately managed by medical interventions

Underlying disorders and contributory factors must be controlled. In particular, controlling hyperglycemia in patients with diabetic nephropathy and controlling hypertension in all patients substantially slows deterioration of GFR.

For **hypertension**, the recommended standardized systolic blood pressure goal (office-based measurement) is < 120 mmHg, when tolerated. However, less intensive goals are set for people with frailty, high risk of falls and fractures, limited life expectancy, or symptomatic postural hypotension (1). Angiotensin-converting enzyme (ACE) inhibitors and angiotensin II receptor blockers (ARBs) decrease the rate of decline in GFR in patients with most causes of chronic kidney disease (CKD), particularly those with proteinuria. Increasing evidence suggests that, compared with either medication alone, combined use of ACE inhibitors and ARBs increases incidence of complications and does not slow decline in renal function, even though combined use does reduce **proteinuria** more (2). Sodium-glucose cotransporter-2 (SGLT2) inhibitors delay progression of proteinuric CKD in patients with or without diabetes, although these medications are contraindicated in patients with type 1 diabetes mellitus (3). Patients with type 2 diabetes mellitus and CKD with albuminuria despite ACE/ARB and SGLT2 may benefit from adding a nonsteroidal mineralocorticoid receptor antagonist (NSMRA) (eg, finerenone) as long as eGFR > 25 and there is no hyperkalemia (1).

Activity need not be restricted, although fatigue and lassitude usually limit a patient's capacity for exercise.

Pruritus may respond to dietary phosphate restriction and phosphate binders if serum phosphate is elevated.

Nutrition

Moderate protein restriction (0.8 g/kg/day) among patients with estimated GFR (eGFR) < 60 mL/min/1.73 m² without **nephrotic syndrome** is recommended in patients without frailty, sarcopenia, or metabolic derangements. Severe (0.3 to 0.4 g/kg/day) protein restriction could be considered for motivated patients under the close supervision of a dietitian, to assure intake of adequate essential amino acids. Many uremic symptoms markedly lessen when protein catabolism and urea generation are reduced. Also, the rate of progression of CKD may slow down. Sufficient carbohydrate and fat are given to meet energy requirements and prevent ketosis. Patients for whom < 0.8 g/kg/day of protein has been prescribed should be closely followed by a dietitian.

Aim for a total dietary sodium intake of < 2 g/day, especially for patients with **volume overload**, **elevated blood pressure**, or **heart failure**.

Because dietary restrictions may reduce necessary vitamin intake, patients should take a multivitamin containing water-soluble vitamins. Administration of vitamins A and E is unnecessary. Vitamins D2 (ergocalciferol) or D3 (cholecalciferol) are not given routinely but are used based on blood levels of vitamin D 25-OH and PTH.

Dyslipidemia should be addressed. Dietary modification may be helpful for hypertriglyceridemia. Statins are effective for hypercholesterolemia. Fibric acid derivatives (clofibrate, gemfibrozil) may increase risk of **rhabdomyolysis** in patients with CKD, especially if taken with statin medications, whereas ezetimibe

(which reduces cholesterol absorption) appears relatively safe. Correction of hypercholesterolemia is intended to reduce risk of cardiovascular disease, which is increased in patients with CKD (4).

Mineral and bone disorders

Based on updated KDIGO (Kidney Disease Improving Global Outcomes) 2024 clinical practice guidelines (1), it is recommended that serum levels of calcium, phosphate, PTH, vitamin D 25-OH, and alkaline phosphatase activity be monitored beginning in CKD stage 3a. Frequency of monitoring depends on severity of CKD, magnitude of above abnormalities, and frequency of therapeutic interventions. Bone biopsy is the most definitive evaluation to determine the type of renal osteodystrophy.

[Hyperphosphatemia](#) should be treated with

- Dietary phosphate restriction
- Phosphate binders

Phosphate restriction to 0.8 to 1 g/day of dietary intake is typically sufficient to normalize serum phosphate level in patients with $\text{eGFR} < 60 \text{ mL/min/1.73 m}^2$. Additional intestinal phosphate binders (calcium-containing [eg, calcium acetate] or non-calcium-containing [eg, sevelamer]) may be necessary for adequate control of hyperphosphatemia, which has been associated with increased cardiovascular risk. Non-calcium-containing binders are preferred in patients with [hypercalcemia](#), suspected adynamic bone disease, or evidence of vascular calcification on imaging. If calcium-containing binders are prescribed, then the total dietary and medication sources of calcium should not exceed 2000 mg/day in patients with $\text{eGFR} < 60 \text{ mL/min/1.73 m}^2$.

[Vitamin D deficiency](#) should be treated with cholecalciferol (vitamin D3) or ergocalciferol (vitamin D2) to target serum vitamin D 25-OH level approximately 30 to 50 ng/mL, as long as there is no hyperphosphatemia or hypercalcemia.

The optimal level of PTH in patients with CKD stages 3a to 5 not on [dialysis](#) is not known. However, if PTH levels are progressively rising or are markedly elevated (above 9 times the upper limit of normal for the assay), despite treatment of hyperphosphatemia and vitamin D deficiency, then an active vitamin D analog (eg, calcitriol) or a calcimimetic (for example, cinacalcet) is recommended. PTH levels are not corrected to normal because doing so risks precipitating adynamic bone disease.

Fluid and electrolytes

Restricted water intake is required only when serum sodium concentration is $< 135 \text{ mmol/L}$ (or mEq/L) or there is heart failure or severe edema.

Sodium restriction of $< 2 \text{ g/day}$ is recommended for CKD patients with $\text{eGFR} < 60 \text{ mL/min/1.73 m}^2$ who have [hypertension](#), [volume overload](#), or [proteinuria](#).

Potassium restriction is individualized based on serum level, eGFR, dietary customs, and use of medications that increase potassium levels (eg, ACE, ARBs, or potassium-sparing diuretics). Typically, potassium restriction is not needed with $\text{eGFR} > 30 \text{ mL/min/1.73 m}^2$. Treatment of mild to moderate [hyperkalemia](#) (5.1 to 6 mmol/L or mEq/L) entails dietary restriction (including avoiding salt substitutes),

correction of metabolic acidosis, and use of potassium-lowering diuretics (eg, furosemide) and gastrointestinal cation exchangers (eg, patiromer). [Severe hyperkalemia](#) (> 6 mmol/L or mEq/L) warrants [urgent treatment](#).

[Metabolic acidosis](#) should be treated to bring serum bicarbonate to > 18 mmol/L to help reverse or slow muscle wasting, bone loss, and progression of CKD. Acidosis can be corrected with oral alkali sources such as sodium bicarbonate or an alkaline-ash diet (primarily fruits and vegetables). Sodium bicarbonate 1 to 2 g orally twice a day is given and the amount is increased gradually until bicarbonate concentration is > 18 mmol/L or until evidence of sodium overloading prevents further therapy. If the alkaline-ash diet is used, serum potassium is monitored because fruits and vegetables contain potassium (5).

Anemia and coagulation disorders

[Anemia](#) is a common complication of moderate to advanced CKD ($\geq$ stage 3). For patients receiving peritoneal or hemodialysis whose hemoglobin is ≤ 9 to 10 g/dL, consider treating with erythropoiesis-stimulating agents (ESA), such as recombinant human erythropoietin (eg, epoetin alfa). For patients not on dialysis, individualize treatment based on patient symptoms and risk-benefit balance (6). Due to risk of cardiovascular complications, including [stroke](#), [thrombosis](#), and death, the lowest dose of these agents needed to keep the Hb between 10 and 11 g/dL is used.

Because of increased iron utilization with stimulated erythropoiesis, iron stores must be replaced, often requiring parenteral iron. Iron concentrations, iron-binding capacity, and ferritin concentrations should be followed closely. Target transferrin saturation (TSAT), calculated by dividing serum iron by total iron binding capacity and multiplying by 100%, should be > 20%. Target ferritin in patients not on dialysis is >100 ng/mL. Transfusion should not be done unless anemia is severe (Hb < 8 g/dL) or causes symptoms.

The bleeding tendency in CKD rarely needs treatment. Cryoprecipitate, red blood cell transfusions, desmopressin 0.3 to 0.4 mcg/kg (20 mcg maximum) in 20 mL of isotonic saline IV over 20 to 30 minutes, or conjugated estrogens 2.5 to 5 mg orally once a day help when needed. The effects of these treatments last 12 to 48 hours, except for conjugated estrogens, which may last for several days.

Heart failure

Symptomatic heart failure is treated with

- Sodium restriction
- [Diuretics](#)
- Sometimes, dialysis

Loop diuretics such as furosemide usually are effective even when renal function is markedly reduced, although large doses may be needed. If left ventricular function is depressed, [ACE inhibitors](#) (or [ARBs](#)) and [beta-blockers](#) (carvedilol or metoprolol) should be used. Aldosterone receptor antagonists (eg, spironolactone) and sometimes sodium-glucose cotransporter-2 (SLGT2) inhibitors are recommended in

patients with advanced stages of heart failure. Digoxin may be added, but the dosage must be reduced based on degree of renal function.

Moderate or severe [hypertension](#) should be treated to avoid its deleterious effects on cardiac and renal function. Patients who do not respond to sodium restriction (1.5 g/day), should receive diuretics. Loop diuretics (eg, furosemide) may be combined with thiazide diuretics (eg, chlorthalidone, hydrochlorothiazide, metolazone) if hypertension or edema is not controlled. Even in renal failure, the combination of a thiazide diuretic with a loop diuretic is quite potent and must be used with caution to avoid overdiuresis.

Occasionally, dialysis may be required to control heart failure. If reduction of the volume of extracellular fluid does not control blood pressure, conventional antihypertensives (ACE inhibitors, ARBs, beta-blockers) are added. Azotemia may increase with such treatment and may be necessary for adequate control of heart failure and/or hypertension.

Medications

Renal excretion of medications is often impaired in patients with renal failure. Common medications that require revised dosing include penicillins, cephalosporins, aminoglycosides, fluoroquinolones, vancomycin, and digoxin. [Hemodialysis](#) reduces the serum concentrations of some medications, which should be supplemented after hemodialysis. It is strongly recommended that physicians consult a reference on drug dosing in renal failure before prescribing medications to these very vulnerable patients ([7-9](#)).

Most experts recommend avoiding NSAIDs (nonsteroidal anti-inflammatory drugs) in patients with CKD because they may worsen renal function, exacerbate hypertension, and precipitate electrolyte disturbances.

Certain medications should be avoided entirely in patients with chronic kidney disease with eGFR < 60 mL/min/1.73m². They include nitrofurantoin and phenazopyridine. The MRI contrast agent gadolinium has been associated with the development of [nephrogenic systemic fibrosis](#) in patients with estimated GFR < 30 mL/min/1.73m² in the past. More recently, class II gadolinium agents are considered safer and preferred when gadolinium is indicated for patients with eGFR <30 or on dialysis ([9](#)).

Dialysis

Dialysis is usually initiated at the onset of either of the following:

- Uremic symptoms (eg, anorexia, nausea, vomiting, weight loss, pericarditis, pleuritis)
- Difficulty controlling fluid overload, hyperkalemia, or acidosis with medications and lifestyle interventions

These problems typically occur when the estimated GFR reaches ≤ 10 mL/min in a patient without diabetes or ≤ 15 mL/min in a patient with diabetes; patients whose estimated GFR values are near these values should be closely monitored so that these signs and symptoms are recognized early. Dialysis is best anticipated so that preparations can be made and urgent insertion of a hemodialysis catheter can be avoided. Such preparations usually begin when the patient is in early to mid stage 4 CKD;

preparation allows time for patient education, selection of the type of dialysis, and timely creation of an [arteriovenous fistula](#) or placement of a [peritoneal dialysis catheter](#). (For dialysis preparation, see [Hemodialysis](#).)

Pearls & Pitfalls

- Begin preparation for dialysis, kidney transplantation, or palliative care during early to mid stage 4 CKD to allow adequate time for patient education and selection of treatment modality, along with any associated preparatory procedures.

Transplantation

If a living kidney donor is available, better long-term outcomes occur when a patient receives the transplanted kidney early, even before beginning dialysis. Patients who are transplant candidates but have no living donor should be placed on the waiting list of their regional transplant center early because wait times may exceed several years in many regions of the United States.

Treatment references

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Prognosis for Chronic Kidney Disease

Progression of chronic kidney disease (CKD) is predicted in most cases by the degree of proteinuria. (See the [Kidney Failure Risk Equation](#).) Patients with nephrotic-range proteinuria (> 3 g/24 h or urine protein/creatinine ratio > 3) usually have a poorer prognosis and progress to renal failure more rapidly. Progression may occur even if the underlying disorder is not active. In patients with urine protein < 1.5 g/24 h, progression usually occurs more slowly if at all. [Hypertension](#), [acidosis](#), and [hyperparathyroidism](#) are associated with more rapid progression as well.

Key Points

- Common causes of CKD in the United States are diabetic nephropathy (the most common), hypertensive nephrosclerosis, glomerulopathies, and metabolic syndrome.
- Effects of CKD can include hypocalcemia, hyperphosphatemia, metabolic acidosis, anemia, secondary hyperparathyroidism, and renal osteodystrophy.
- Distinguish CKD from acute kidney injury based on history, clinical findings, routine laboratory tests, and ultrasound.
- Control underlying disorders (eg, diabetes) and BP levels (usually with an ACE inhibitor or ARB).
- Treat patients with proteinuric CKD with an ACE inhibitor or ARB plus an SGLT2 inhibitor and then a NSMRA.
- Give supplemental vitamin D and/or sodium bicarbonate and restrict potassium and phosphate as needed.
- Treat heart failure, anemia, and other complications.
- Educate patients with advanced CKD on treatment options (dialysis, kidney transplantation, or palliative care) early, to allow adequate time for planning.
- Initiate dialysis for patients with severely decreased eGFR when signs and symptoms are inadequately controlled with medications and lifestyle interventions.



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